

1. Course number: 010153103
Course name: Algorithms and Data Structures
2. Credits: 3(3-0-6)
Contact hours: 3 hours per week
3. Instructor:
Asst. Prof. Dr. Ruslee Sutthaweekul
4. Textbook:
 - a. Goodrich, Michael T., Roberto Tamassia, and Michael H. Goldwasser. Data structures and algorithms in Java, 6th Edition, John Wiley & Sons, 2014.
5. Specific course information:
 - a. Course Description:
This course requires students to understand basic data structures for computer programming development; list, stack, queue, and tree; algorithms; sorting; searching; recursion.
 - b. Prerequisites: 010153002 Computer Programming
 - c. Course type: Required
6. Specific goals for the course (Course Learning Outcome: CLO)
 - a. Specific outcomes of instruction:
 1. Demonstrate a good understanding of linear data structures, including arrays, linked lists, stacks, and queues.
 2. Implement abstract data types effectively.
 3. Understand non-linear data structures, including trees and graphs.
 4. Select and use appropriate data structures to solve specific problems.
 5. Understand and implement sorting, searching, and hashing algorithms.
 6. Choose and apply appropriate algorithms to solve specific problems.
 - b. Student Outcomes Addressed:

ABET Student Outcome (SO)	Course Learning Outcome: CLO
1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	1-6
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and	4, 6

ABET Student Outcome (SO)	Course Learning Outcome: CLO
welfare, as well as global, cultural, social, environmental, and economic factors	
3. an ability to communicate effectively with a range of audiences	-
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	-
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	-
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	2, 4, 6
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies	4, 6

7. Course Learning Outcomes and Assessment tools:

Course Learning Outcome: CLO	SO	Assessment tools
1. Demonstrate a good understanding of linear data structures, including arrays, linked lists, stacks, and queues.	1	Exams
2. Implement abstract data types effectively.	1, 6	Exams, Assignment
3. Understand non-linear data structures, including trees and graphs.	1	Exams
4. Select and use appropriate data structures to solve specific problems.	1, 2, 6, 7	Exams, Assignment
5. Understand and implement sorting, searching, and hashing algorithms.	1	Exams
6. Choose and apply appropriate algorithms to solve specific problems.	1, 2, 6, 7	Exams, Assignment

8. Grading Plan:

81 – 100	= A	51 – 60	= C
76 – 80	= B+	46 – 50	= D+
71 – 75	= B	41 – 45	= D
61 – 70	= C+	0 – 40	= F

9. Brief list of topics to be covered

1. Introduction to ADS and Programming Review
2. Java & Object Oriented Design
3. Algorithm Analysis
4. Array Lists
5. Linked Lists
6. Stacks, Queue
7. Recursion
8. Trees, Binary Trees
9. Graphs
10. Sorting, Searching
11. Hashing